# MINI-WARMAN DESIGN CONCEPT

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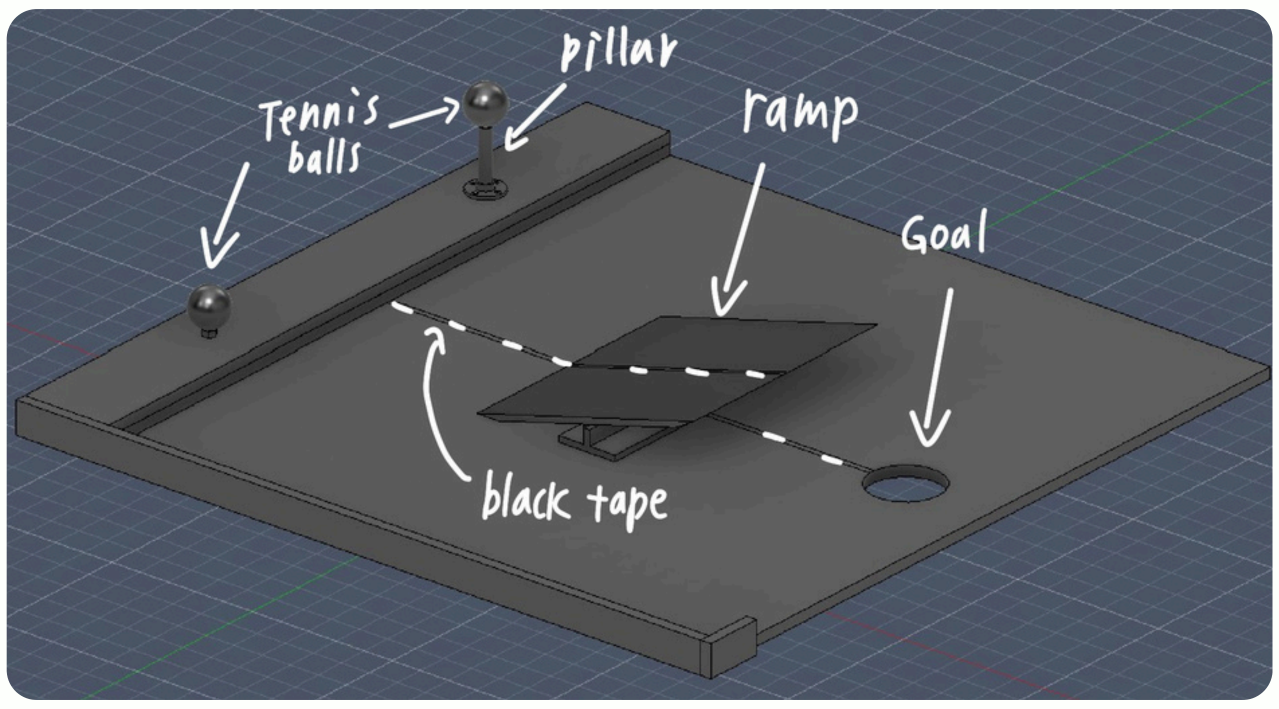
## THE COURSE



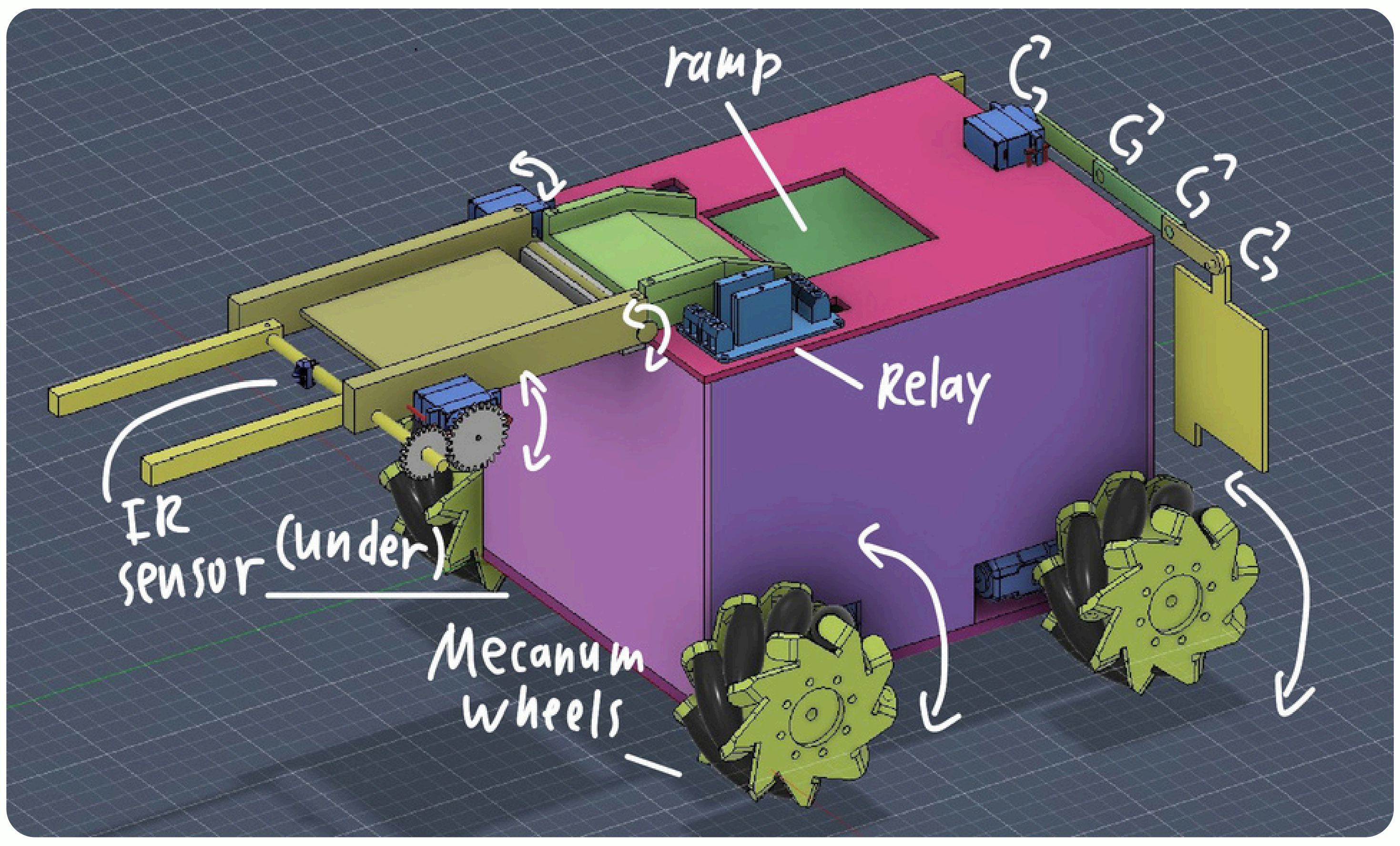

### The problem

Transporting tennis balls to a designated point in an unique obstacle course.

### Solution

Autonomous robot that collect, store, and release payloads.

## THE ROBOT - MAIN FEATURES




### Servo controlled arms

Made of a bottom segment and top segment. Used to pick up balls.

### Ramp storage + Trap door

Ramp within the robot can store the ball. Back panel can then open to release it.

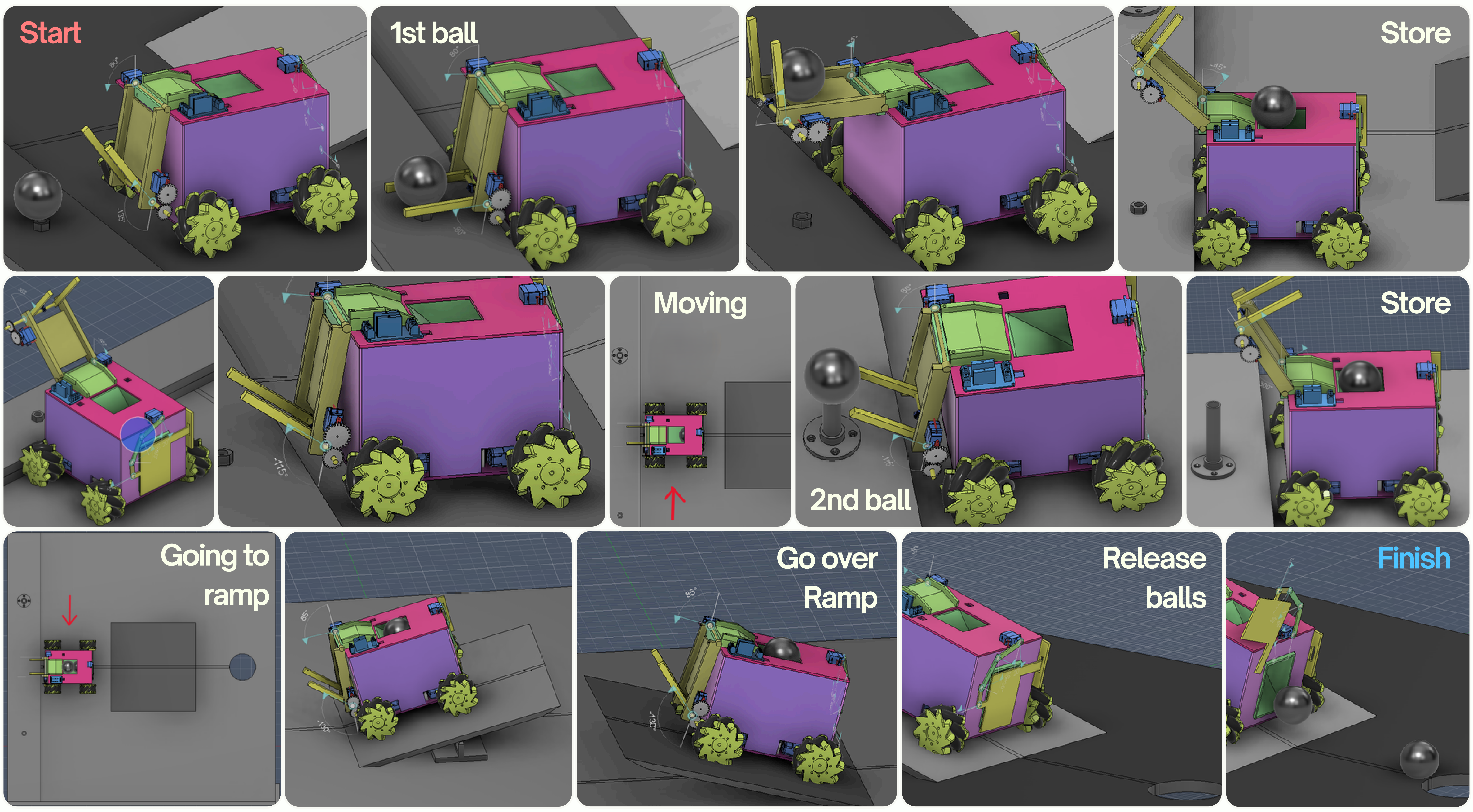
### Mecanum wheels

Mecanum wheels allow movement in all directions.

### IR sensors + Relay

Helps robot find balls, and locate where the ramp is via the tape on the course.

## OPERATION



# WHY THIS DESIGN?

## CHEAP

### Relatively few parts
Doesn't require many extra pieces to create body & arm.

### High tolerance
Doesn't require high degree of precision to work.

### Fast production
Most of the robot can be laser cut.

### Easy assembly
No complicated part fittings, all can be glued or screwed in.

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## RELIABLE

### Robust
Adjusts behavior using IR sensors in real time.

### Low center of gravity
Heavy towards the bottom due to battery placement and ball storage.
= Higher stability at high speeds
= Easily flips over ramp.

### Easily redeployable
Simply place back at the starting point to reset.

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## FAST

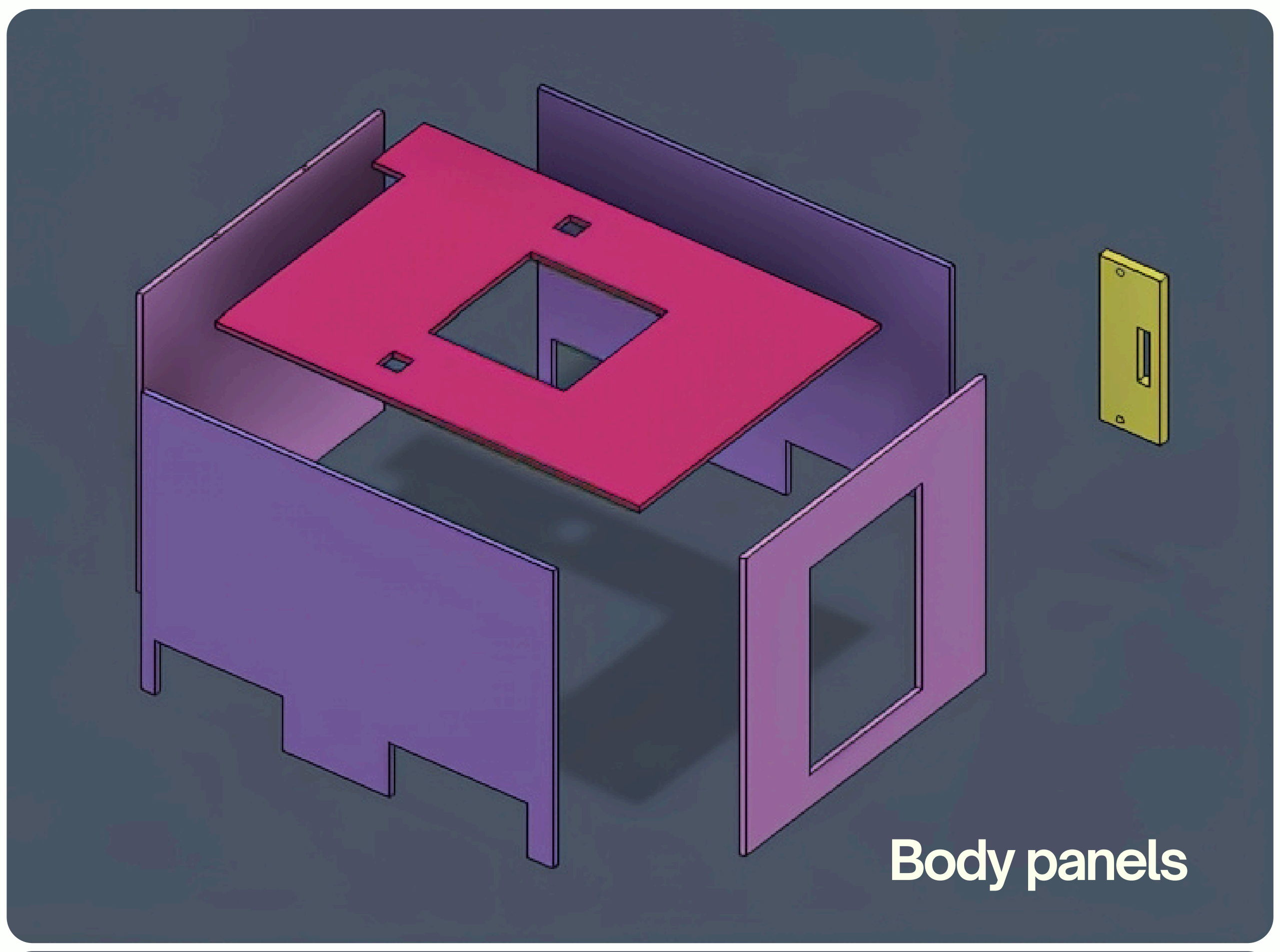
### No time on turning
Mecanum wheels mean no time wasted turning.

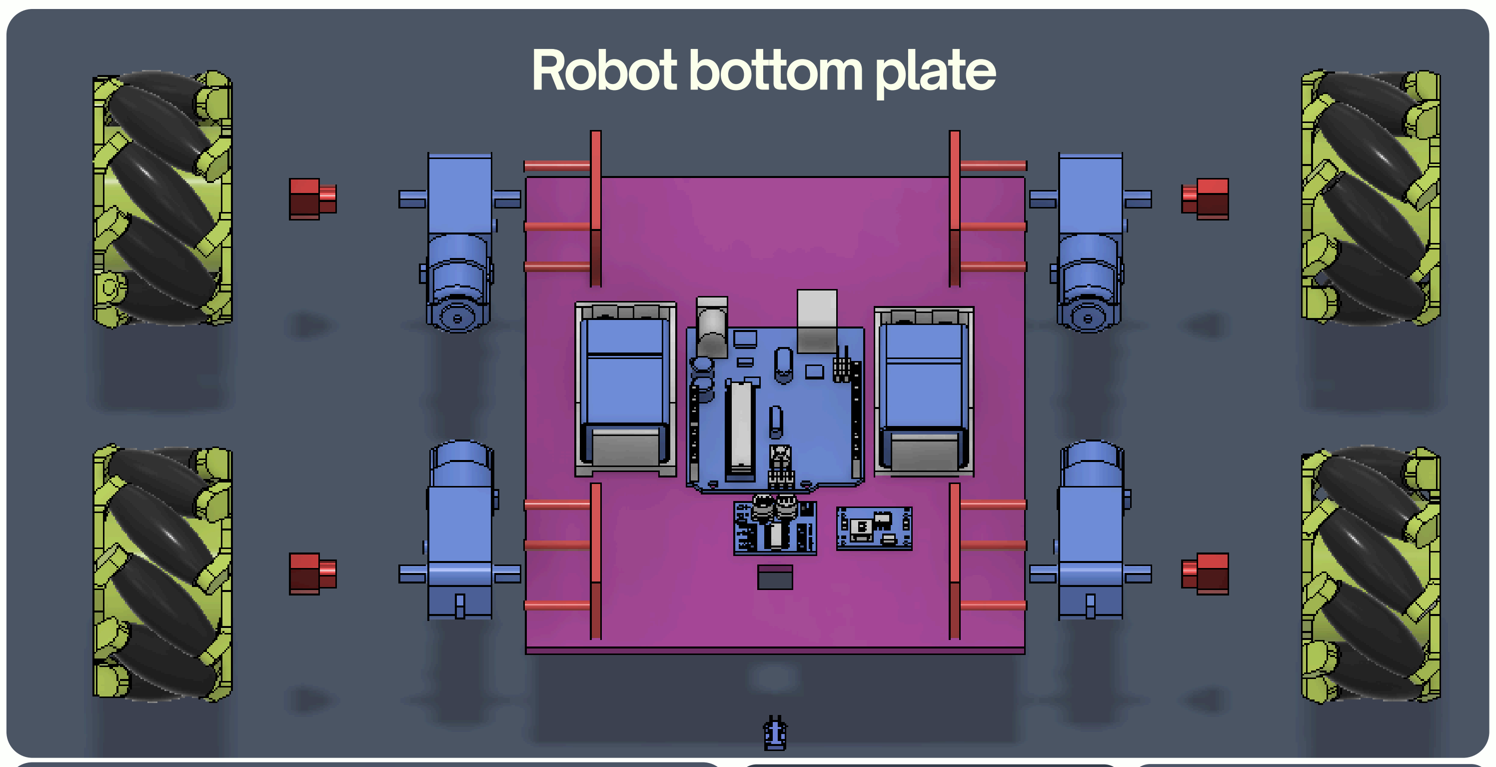
### Small profile
Has small size and relatively low weight.

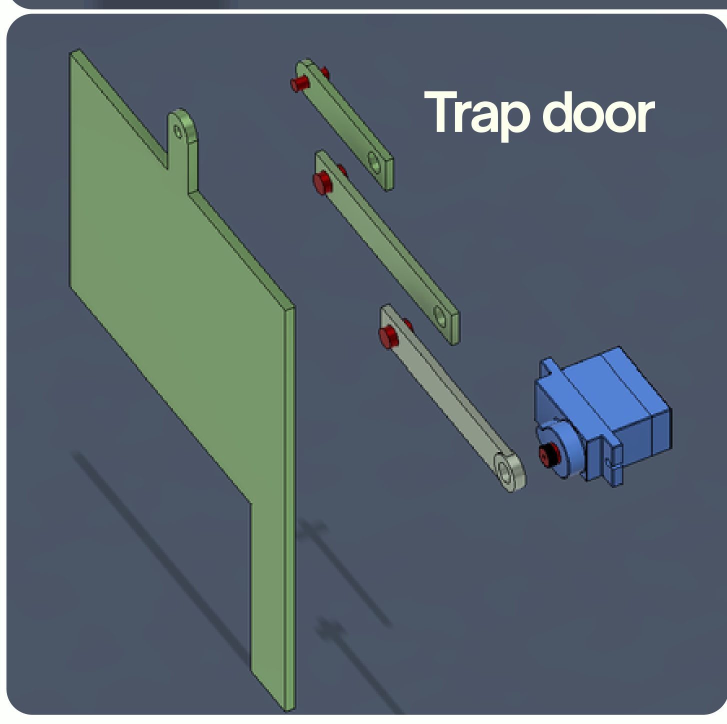
### Assisted by newton
Utilising the ramp means faster transport time to the goal.

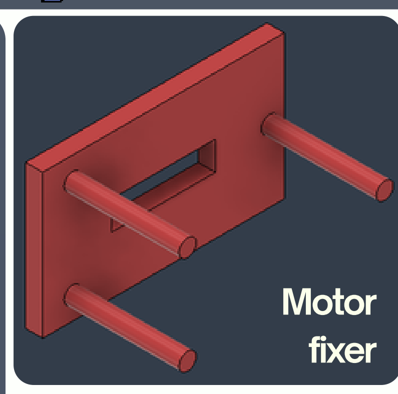
### Speed & Precision
Arms controlled by servos = precise rotations.

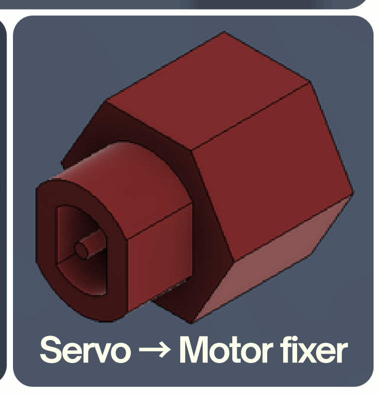
## ASSEMBLY

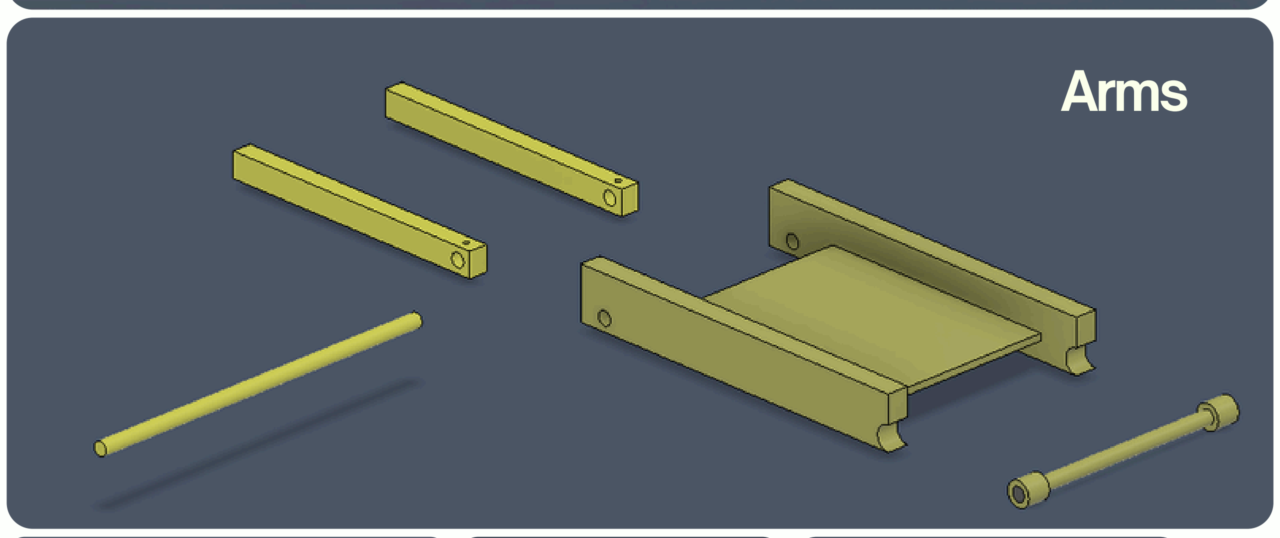


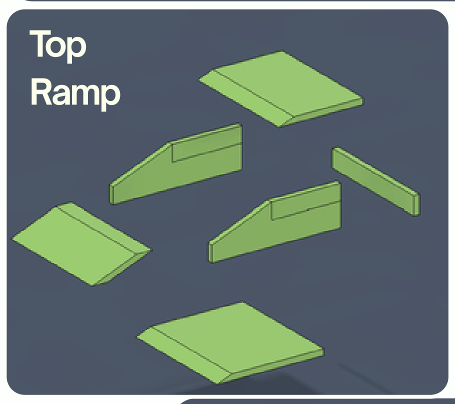



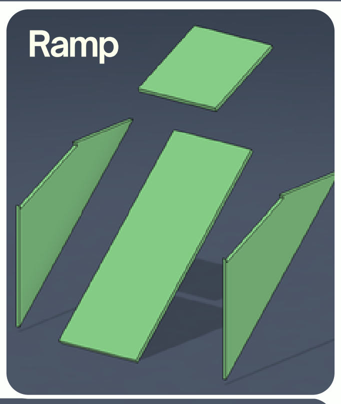



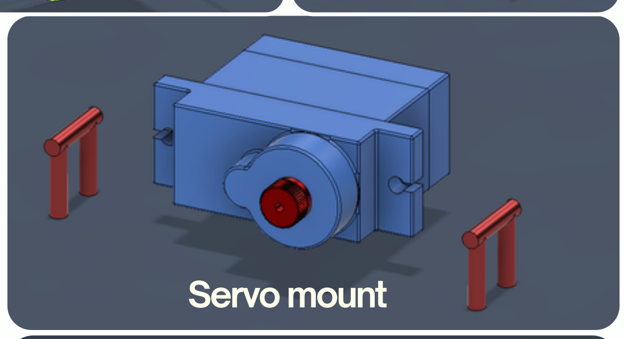

Fasteners

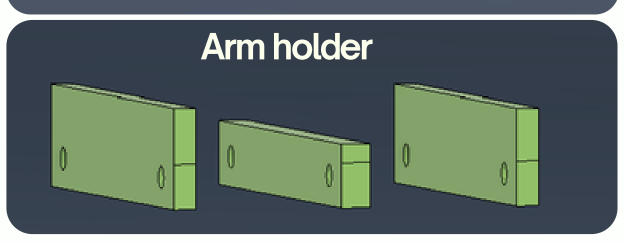
Electronics








Parts

Robot main body

Thanks for considering my design!